

Week 8 (LAB4): Factorial Design

PSTAT122: Design and Analysis of Experiments

Winter 2026

STUDENT NAME

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Due Date

Due Date: Friday, February 27, 2026, 11:59 PM

1 Methods

- Start with a brief description of the aim of this lab The aim of this lab is to design a 2^2 factorial experiment that collects data to investigate how factor A, jumping jacks before hold breath, and factor B, arms up while taking a breath, affect how long you can hold your breath. We randomly assign the control, just normally holding your breath, arms up while taking a breath, jumping backs before holding your breath, and doing both to collect data for this experiment.
- Then give a description of how your data were collected, including how randomization was performed Randomization was performed by the ta, utilising set seed. Here is a sample of the code used for randomization

```
set.seed(1225) # for reproducibility
treatments <- rep(c("control", "arms up", "jumping jacks", "both"), each = 5)
sample(treatments)
```

```
[1] "arms up"      "control"      "both"         "both"
[5] "both"        "arms up"     "control"      "jumping jacks"
[9] "arms up"     "arms up"     "control"      "both"
[13] "arms up"     "jumping jacks" "control"      "jumping jacks"
[17] "control"     "jumping jacks" "both"         "jumping jacks"
```

Data was collected by recording time holding breath recorded at whatever level of precision each individuals timing device gives. All times where compiled into an excel file. The data was ensured to be balanced by pre-assigning an equal number of students to each of the four treatment combinations. Since we had 24 students, our design is a nearly balanced design.

- Describe any technical issues that may have come up, such as timer problems, or anything else.

One drawback is that we where unable to record the order of data collection. Also, because when we first did the set seed randomization there was only 20 participants, the additional four participants where not part of this initial randomization which reduced the accuracy of our experiment. Another drawback is that by each timing ourselves, we add the nuisance factor of each individual has a different accuracy with the timing that is not consistent between each of us.

2 Results

- Include an image of the whiteboard data. If you have saved the image file in the same directory as your `.Rmd` file, then you can include the image into your report with:
- Carefully write out your null and alternative hypotheses for your primary question of interest; if you use any mathematical symbols, carefully define what they represent.

Factor A (Exercise) Null: There is no difference in the mean breath holding time between students who did jumpingjack and students in the control Alternative: Performing jumpin jacks signifiacntly changes mean breath holding time

Factor B (Arms Up) Null: There is no difference in the mean breath-holding time between students who held their arms up and students who kept their arms in a normal position Alternative: Raising ones are significantly changes the mean breath holding time

Interaction of Exercise and Arms Up Null: There is no interaction between exercise and posture. The effect of jumping jacks on breath holding time is the same regardless of whether the arms are up or in a normal position Alternative:There is a significant interaction between exercise and arms up. The effect of jumping jacks on breath holding time depends on the position of the arms.

```
library(tidyverse)
library(readxl)
library(knitr)

data <- read_excel("Lab4Data.xlsx", col_names = FALSE)
colnames(data) <- c("Name", "Treatment", "Time")

# Factors for x axis
data_factors <- data %>%
  mutate(
    Exercise = ifelse(Treatment %in% c("jumping jacks", "both"), "JJ", "No JJ"),
    ArmsUp = ifelse(Treatment %in% c("arms up", "both"), "Arms Up", "Normal")
  ) %>%
  mutate(Exercise = factor(Exercise), ArmsUp = factor(ArmsUp))
kable(data_factors,
      col.names = c("Name", "Original Treatment", "Time (s)", "Exercise Factor", "Posture Factor"),
      caption = "Experimental Dataset for Breath-Holding Study")
```

Table 1: Experimental Dataset for Breath-Holding Study

Name	Original Treatment	Time (s)	Exercise Factor	Posture Factor
Grace Bianchi	arms up	36.32	No JJ	Arms Up
Lily Nishizawa	control	46.05	No JJ	Normal
Atia Yasmin	both	20.33	JJ	Arms Up
brendan chung	both	40.67	JJ	Arms Up
Raiyan khan	both	33.62	JJ	Arms Up
Ryan Fabrick	arms up	36.67	No JJ	Arms Up
Ethan Alteza	control	54.32	No JJ	Normal
Gursimrat Grewal	jumping jacks	32.64	JJ	Normal
Chloe Lo Papa	arms up	41.72	No JJ	Arms Up
Ryan Chang	arms up	39.66	No JJ	Arms Up
Daniel Larson	control	121.93	No JJ	Normal
Ethan Chuang	both	34.71	JJ	Arms Up
Aakash Senthilnathan	arms up	101.20	No JJ	Arms Up
Maya Trieu	jumping jacks	25.10	JJ	Normal
Bernard Abousleiman	control	31.95	No JJ	Normal
Saketh Ayinaparthi	jumping jacks	27.64	JJ	Normal

Name	Original Treatment	Time (s)	Exercise Factor	Posture Factor
Yoshihito Abe	control	51.60	No JJ	Normal
Anhong Tang	jumping jacks	21.35	JJ	Normal
Jerry Yan	both	34.28	JJ	Arms Up
Andres Ulloa	jumping jacks	33.88	JJ	Normal
Alisha Thomas	arms up	28.78	No JJ	Arms Up
Zachary Scholnick	jumping jacks	34.12	JJ	Normal
Nijun Xuan	control	42.05	No JJ	Normal
Jiayao Tao	control	32.05	No JJ	Normal

```

# Summary table
summary_table <- data_factors %>%
  group_by(Exercise, ArmsUp) %>%
  summarize(
    n = n(),
    Mean = mean(Time),
    SD = sd(Time),
    .groups = "drop"
  )
kable(summary_table, digits = 2, caption = "Summary Statistics by Treatment Combination")

```

Table 2: Summary Statistics by Treatment Combination

Exercise	ArmsUp	n	Mean	SD
JJ	Arms Up	5	32.72	7.48
JJ	Normal	6	29.12	5.27
No JJ	Arms Up	6	47.39	26.73
No JJ	Normal	7	54.28	31.07

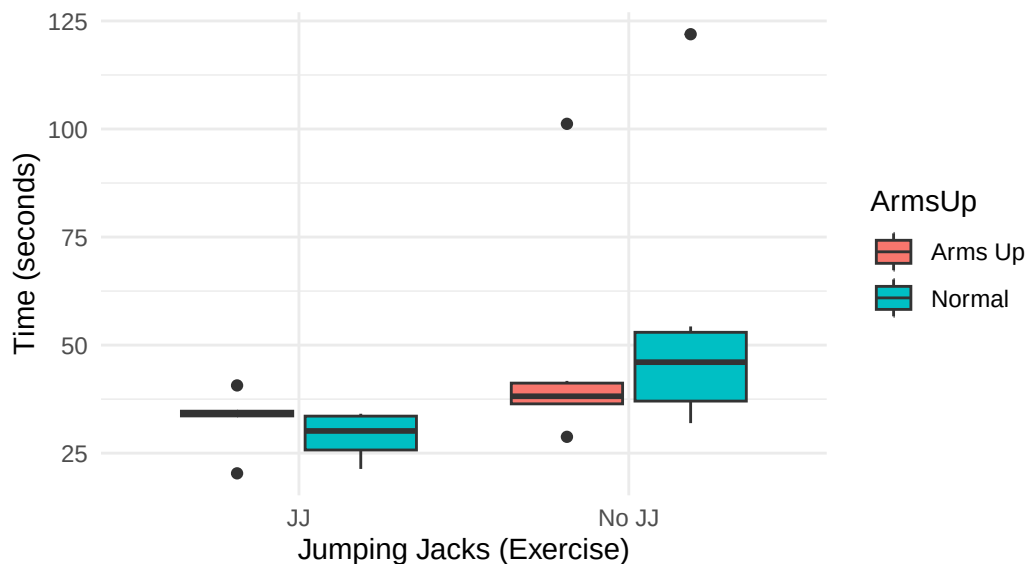
```

# gg plot
ggplot(data = data_factors, mapping = aes(x = Exercise, y = Time, fill = ArmsUp)) + geom_boxplot() +
  labs(title = "Breath-Holding Time: Exercise vs. Arms Up",
    subtitle = "PSTAT 122 Lab 4 Data",
    x = "Jumping Jacks (Exercise)",
    y = "Time (seconds)") +
  theme_minimal()

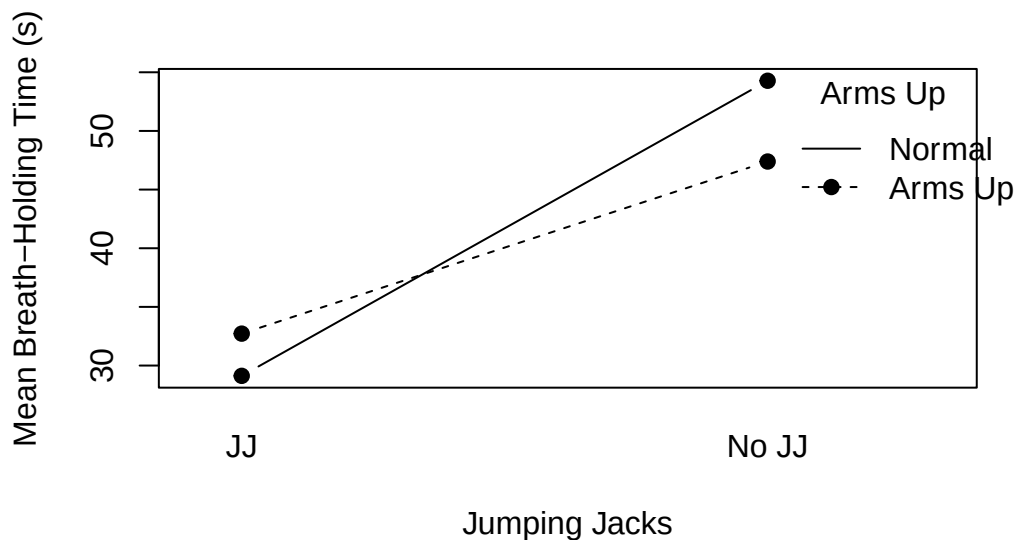
```

Breath-Holding Time: Exercise vs. Arms Up

PSTAT 122 Lab 4 Data



```
# Interaction Plot
interaction.plot(x.factor = data_factors$Exercise,
  trace.factor = data_factors$ArmsUp,
  response = data_factors$Time,
  type = "b",
  pch = 19,
  xlab = "Jumping Jacks",
  ylab = "Mean Breath-Holding Time (s)",
  trace.label = "Arms Up")
```



```
model <- aov(Time ~ Exercise * ArmsUp, data = data_factors)
summary(model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Exercise	1	2465	2465.5	5.069	0.0358 *
ArmsUp	1	26	25.9	0.053	0.8197

Exercise:ArmsUp 1 163 162.6 0.334 0.5695

Residuals 20 9728 486.4

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Table 1 gives all of the data we collected in lab. Table 2 describes the Mean of each of the four treatments breath holding time. It also gives the standard deviation. From this table, we can see that jumping jacks cause a much lower breath holding time in comparison to no jumping jacks. The GG plot Breath-Holding Time shows the spread of the data. We can see that NO JJ boxes are much higher on the y axis, meaning that the individual could hold their breath longer. It seems to show that students that did jumping jacks held their breath for a significantly shorter time. From this graph we also see that holding ones arms up does really change breath time. Arms up has a consisent breath time to the jj and normal plot. The interaction plot has both lines basically parallel. This suggusts there is no interaction between jumping jacks and arms up changing breath time. It also shows that the main effect of exercise and that jumung jacks lower mean beathtimes. It also shows that the main effect of arms up is small. Now looking at anova. The steep line in the interaction plot correlates to the significant p-value for exercise. We can conclude that jumping jacks do significantly affect breath time. Additionally, our anova plot confirms that arms up does not significantly affect the breath time as the p value is .8197, this non significant result relates to the lines being close together in the interaction plot. And, the p-value for interaction is 0.5695 which is not significant indicating there is no significant interaction between the two factors. This means that the effect of jumping jacks on breath-holding time is independent of arms up.

3 Discussion

The aim of this lab is to design a 2^2 factorial experiment that collects data to investigate how factor A, jumping jacks before hold breath, and factor B, arms up while taking a breath, affect how long you can hold your breath. We randomly assign the control, just normally holding your breath, arms up while taking a breath, jumping backs before holding your breath, and doing both to collect data for this experiment. In our lab, we find that jumping jacks do significantly lower the amount of time an individual is able to hold their breath. We found that arms up and the interaction between our two factors are insignificant. Our randomization was performed using `set.seed(1225)` for the first 20 participants, but the final four participants were added afterward without a new randomization process. This is a drawback because of selection bias, our final four students were not truly randomized into their groups. There is also a dely between a person stopping their breath and physicall pressing stop on the phone and this amount of time varies by person to person. Also a person might cheat by not holding their breath for as long as they can, or dely how wuickly they stop the timer. This design also lacked blocking as a person who exercises frequently and swims might have a higher lung capacity or be used to holding their breath for long times which could account for the large outliers we saw.